



NATIONAL TEACHERS COLLEGE

“AquaTrace: Community Water Access and Monitoring System”

A project presented to the faculty of the
School of Arts, Science and Technology
National Teachers College
Quiapo, Manila

In Partial Fulfillment of the Requirements
in IT Elective 1
for the Degree of
Bachelor of Science in Information Technology

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May 22, 2026

1. **Introduction**

AquaTrace is a desktop based Windows Forms information system developed using VB.NET and .Net 10 to support the United Nations Sustainable Development Goal 6 (SDG 6), Clean Water and Sanitation. The system is designed to assist local government units, barangay officials, and water district personnel in efficiently monitoring, managing and maintaining clean water access points across communities in the Philippines.

Access to clean and safe water is a fundamental human right, yet many communities, particularly in rural and semi urban areas of the Philippines, continue to struggle with insufficient water supply management. The lack of a centralized, digital system for tracking water source conditions, maintenance schedules and household coverage often leads to delayed responses, unresolved issues, and poor resource allocation.

AquaTrace addresses this gap by providing a structured, user-friendly platform that digitalizes the entire water source management workflow, from recording new water access points and tracking their health status, to logging maintenance activities and generating summary reports for decision makers. The system incorporates role-based access control, ensuring that data integrity is maintained while allowing both administrators and field to perform their respective tasks.

2. **Problem Statement**

Despite the growing need for clean water access monitoring in Philippine communities, most barangay and municipal water management offices continue to rely on manual, paper based record keeping. This approach presents several critical problems.

- No centralized database - water source records are scattered across physical logbooks, spreadsheets and individual files, making data retrieval slow and error prone.

- Missed maintenance schedules - without a structured logging system, maintenance activities are often documented or forgotten, leading to degrading water sources.
- Unknown household coverage - officials lack accurate data on how many households each water source serves, making resources planning difficult.
- Lack of real-time status monitoring - identifying water sources that are inactive or critically damaged requires manual field inspection with no digital reference.
- Absence of reporting tools - generating summary reports for higher authorities requires manual compilation, which is time consuming and prone to errors.
- Uncontrolled data access - without a login system, sensitive water infrastructure data can be accessed or modified by unauthorized personnel.

3. Objectives

General Objective

To design and develop AquaTrace, which provides local government units and water district personnel with a centralized, digital system for monitoring, managing and reporting on community water access points in support of UN SDG 6.

Specific Objectives

1. To implement a secure role-based login system that distinguishes between Admin and Standard User access levels.
2. To develop a Water Sources Management module that allows users to perform full CRUD operations on water access point records.
3. To build a Maintenance Log module that enables personnel to document all maintenance activities linked to specific water sources via Foreign Key.
4. To implement real-time input validation using `String.IsNullOrEmpty()`, `Double.TryParse()`, and `Regex` to ensure data accuracy and system robustness.
5. To integrate a Report Viewer that generates professional summary reports on water source conditions and maintenance history.

6. To apply Object-Oriented Programming principles through dedicated class files with Private Fields and Public Properties (Encapsulation)
7. To design a relational SQL Server LocalDB database with at least two related tables connected via Foreign Key constraints.
8. To separate the application into a Presentation Layer (Forms), Business Logic Layer (Classes), and Data Access Layer (DBManager) following modular architecture principles.

4. **SDG Alignment**

AquaTrace is directly aligned with United Nations Sustainable Development Goal 6: Clean Water and Sanitation, which aims to ensure availability and sustainable management of water and sanitation for all by 2030. The system contributes to this goal by digitizing and centralizing water source management at the community level, enabling faster response to water access issues and better planning for water infrastructure maintenance.

SDG Alignment Table

SDG 6 Target	UN Target Description	AquaTrace Contribution
6.1	Achieve universal equitable access to safe and affordable drinking water for all,	Tracks all community water access points and their coverage in terms of households served, enabling officials to identify underserved areas.
6.3	Improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals.	Records health status and condition of each water source, flagging those needing repair or inspection to prevent water contamination.
6.4	Substantly increase	Monitors water source

	water-use efficiency across all sectors.	capacity and household coverage data, enabling better allocation and utilization of water resources.
6.5	Implement integrated water resources management at all levels.	Provides a centralized digital platform for managing water source data, maintenance logs, and generating reports for planning.
6.B	Support and strengthen participation of local communities in improving water management.	Role-based access allows barangay personnel and admins to collaboratively manage and monitor water source data at the community level.

5. Scope & Limitations

Scope

- User authentication and role-based access control (Admin and Standard User roles).
- Full CRUD operations for Water Sources - adding, viewing, updating, and deleting records.
- Full CRUD operations for Maintenance Logs - recording and managing maintenance activities tied to specific water sources.
- Live search and filtering of water source records by name, location, or status.
- Report generation using Microsoft Report Viewer to produce professional summary reports.
- Relational database with two related tables (WaterSources and MaintenanceLogs) connected via Foreign Key.

- Input validation for all user inputs including required fields, numeric values, and text format checks.

Limitations

- The system is a desktop application and does not support web-based or mobile access.
- Real-time data synchronization across multiple workstations is not supported.
- The system does not include GPS or map integration for visualizing water source locations geographically.
- Password encryption is not implemented in this version - passwords are stored as plain text.
- The system does not send automated notifications for overdue maintenance schedules.

6. Requirements Analysis

Functional Requirements (FR)

FR#	Feature	Description
FR-01	User Login	Users log in with username and password. The system differentiates between Admin and Standard User roles and restricts access accordingly.
FR-02	Water Source CRUD	Full Create, Read, Update, Delete operations on water source records using parameterized queries to prevent SQL Injection.

FR-03	Maintenance Log CRUD	Record, view, update, and delete maintenance activities linked to water sources via Foreign Key relationship on SourceID.
FR-04	Live Search	Filter water source records in real time using SQL LIKE query based on name or location input from the search TextBox.
FR-05	Report Generation	Generate professional summary reports using Microsoft Report Viewer in RDLC format showing water source and maintenance data.
FR-06	Input Validation	All fields validated, required fields via IsNullOrWhiteSpace(), numeric fields via Double.TryParse(), wrapped in Try-Catch-Finally.
FR-07	Role-Based Access	Admin users access all modules including delete operations. Standard users can only view and add records.

FR-08	Dashboard Summary	Dashboard displays total water sources, active count, inactive count, and sources needing repair at a glance.
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Non-Functional Requirements (NFR)

Category	Requirement
UI Consistency (NFR-01)	All controls follow standard naming conventions (btn, lbl, txt, cbo, dgv) with a clean, intuitive layout.
Maintainability (NFR-02)	Code is documented with XML comments and standard green-line comments explaining complex logic in all class files.
Reliability (NFR-03)	Database connection strings managed through a dedicated ConnectionClass, not hardcoded directly in form files.
Robustness (NFR-04)	All database operations wrapped in Try-Catch-Finally blocks with structured error messaging via MessageBox.
Security (NFR-05)	Delete operations and sensitive modules are restricted to Admin role users only to protect data integrity.

7. Tools & Technologies

Category	Tool/ Technology	Purpose
Programming Language	VB.Net	Primary language for Windows Forms desktop

		development.
Framework	.NET 10	Application runtime and class library framework.
IDE	Visual Studio 2026	Development environment for coding, designing and debugging.
Database	Microsoft SQL Server	Stores AquaTrace_DB with WaterSources, MaintenanceLogs, and Users tables
Data Access	Microsoft.Data.SqlClient (ADO.NET)	Uses SqlConnection, SqlCommand, and SqlDataAdapter for parameterized CRUD queries.
Reporting	Microsoft Report Viewer	Generates professional RDLC-based summary reports.
Version Control	GitHub	Source code management and group collaboration repository.
AI Assistant	Claude AI	Code assistance, debugging

8. Database Design

AquaTrace uses a SQL server LocalDB database named AquaTrace_DB containing three related tables. The WaterSources and MaintenanceLogs are connected via a Foreign Key on SourceID. The Users table manages authentication data separately for the login system.

Table 1: WaterSources

Column Name	Data Type	Constraint	Description
SourceID	INT	PK, IDENTITY (1,1)	Auto-increment primary key.
SourceName	NVARCHAR (100)	NOT NULL	Name of the water access point.
Location	NVARCHAR (150)	NOT NULL	Barangay or area location.
SourceType	NVARCHAR (50)	NOT NULL	Well, Pump, Reservoir or Pipeline.
Status	NVARCHAR (50)	NOT NULL	Active, Needs Repair or Inactive.
CoverageHH	INT	NOT NULL	Number of households covered.
LastInspected	DATETIME	NOT NULL	Date of last field inspection.

Table 2: MaintenanceLogs

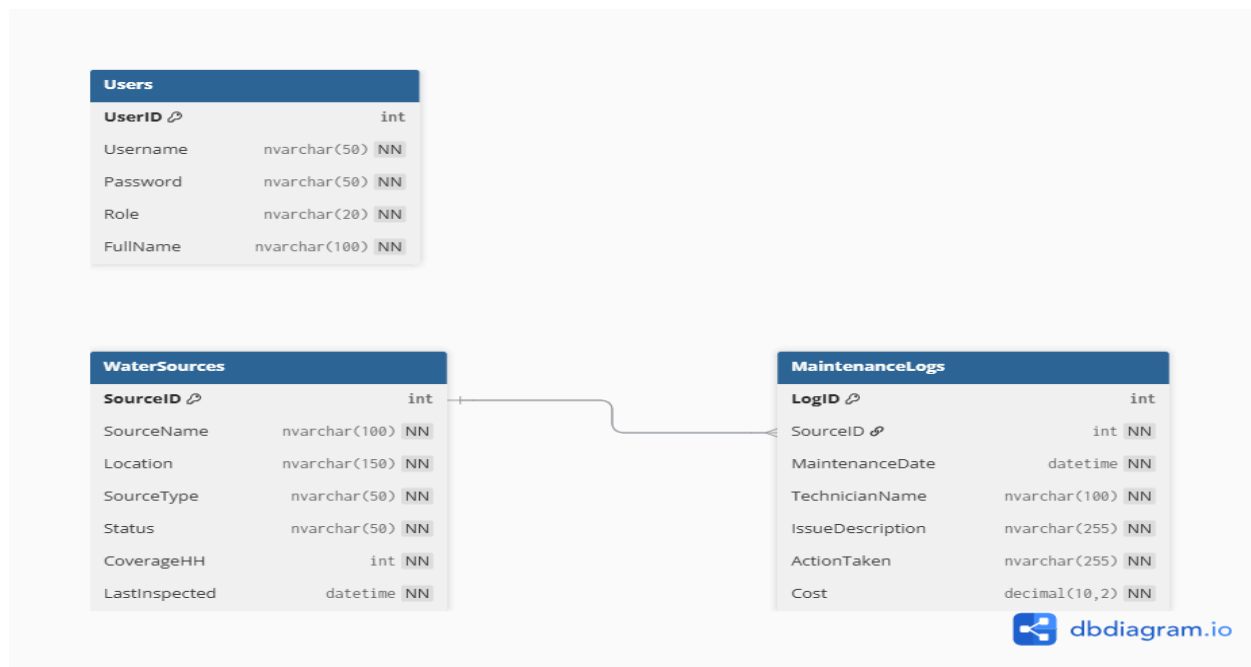
Column Name	Data Type	Constraint	Description
LogID	INT	PK, IDENTITY (1,1)	Auto-increment primary key.
SourceID	INT	FK → WaterSources	Foreign key linking to water source.
MaintenanceDate	DATETIME	NOT NULL	Date when maintenance was performed.
TechnicianName	NVARCHAR (100)	NOT NULL	Name of assigned technician.
IssueDescription	NVARCHAR (255)	NOT NULL	Description of the issue found.
ActionTaken	NVARCHAR (255)	NOT NULL	Maintenance actions performed.
Cost	DECIMAL (10,2)	NOT NULL	Total cost of the maintenance activity.

Table 3: Users

Column Name	Data Type	Constraint	Description
UserID	INT	PK, IDENTITY (1,1)	Auto-increment primary key.
Username	NVARCHAR (50)	NOT NULL	Login username.

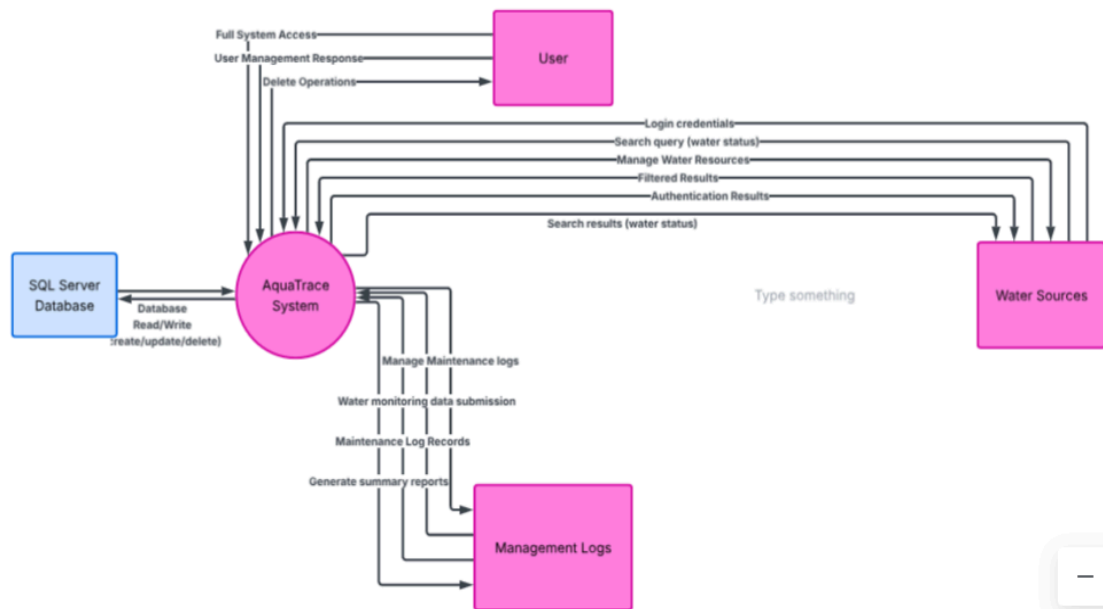
Password	NVARCHAR (50)	NOT NULL	Login password.
Role	NVARCHAR (20)	NOT NULL	Admin or Standard user.
FullName	NVARCHAR (100)	NOT NULL	Full name of the user.

Entity Relationship Diagram (ERD)

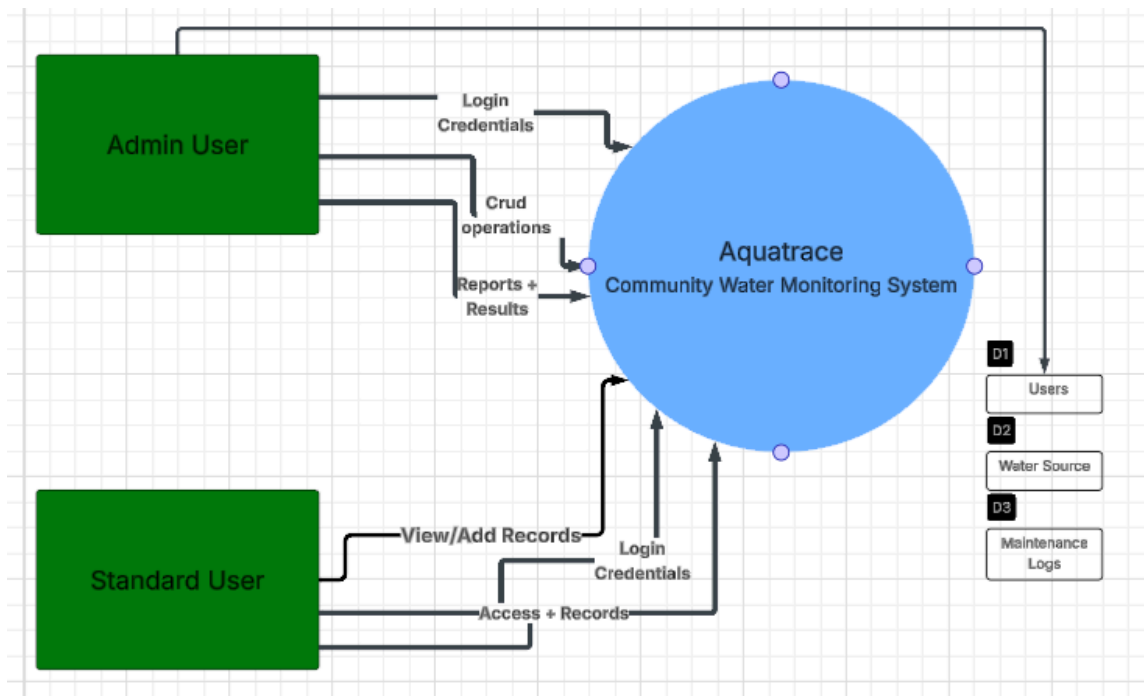


9. System Architecture

DFD Level 0



DFD Level 1

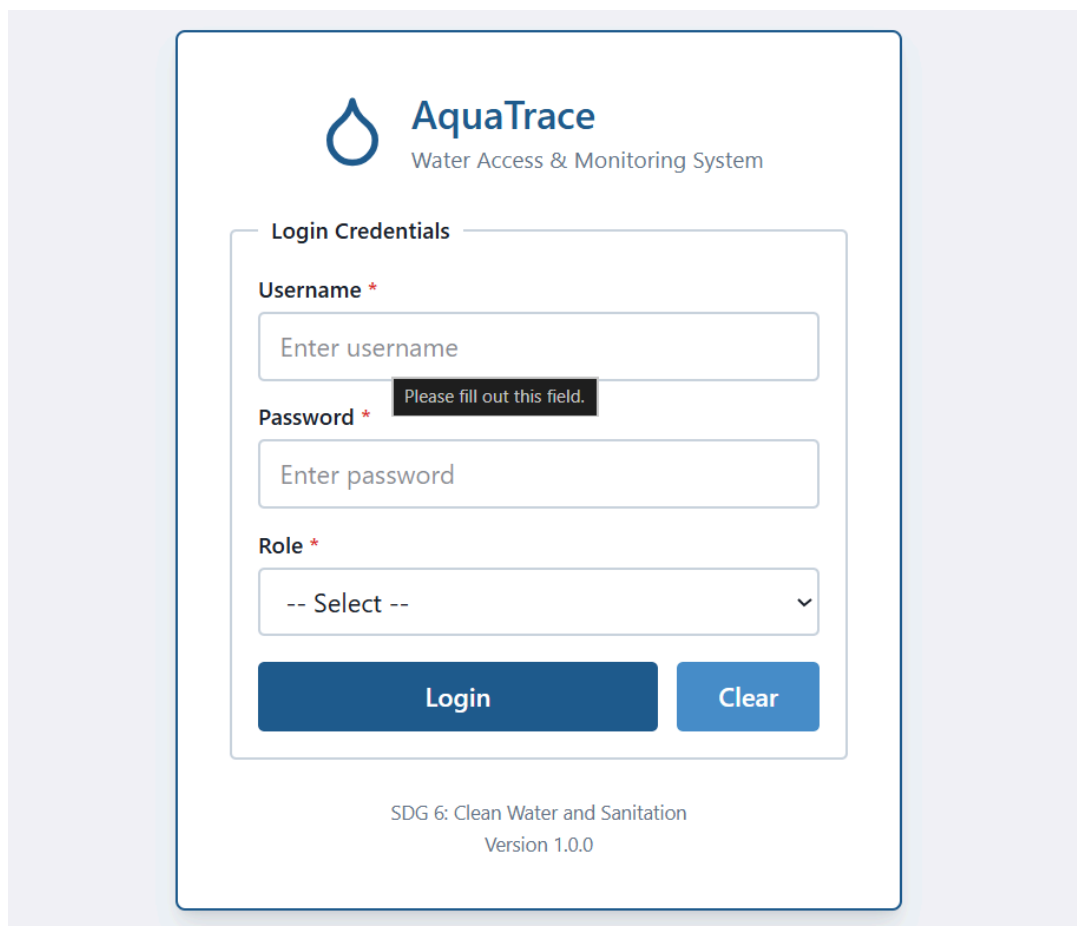


10. UI/UX Design

The AquaTrace interface follows professional UI/UX standards with consistent naming conventions (btn, lbl, txt, cbo, dgv), clean layouts, and intuitive navigation. Controls are organized within GroupBoxes for logical grouping. The blue color scheme reflects the water and clean water theme of SDG 6.

Login Screen

The entry point of the system where users authenticate themselves. Users enter their username, password, and select their role. This ensures role-based access control for the water monitoring system.



The image shows a login screen for the AquaTrace Water Access & Monitoring System. The interface is clean and professional, featuring a blue color scheme. At the top, the AquaTrace logo (a blue water drop) is displayed next to the system name. Below the logo, the text "Water Access & Monitoring System" is visible. The main section is titled "Login Credentials" and contains three input fields: "Username *", "Password *", and "Role *". The "Username" field has a placeholder text "Enter username". The "Password" field has a placeholder text "Enter password" and a tooltip that says "Please fill out this field." The "Role" field is a dropdown menu with the text "-- Select --" and a downward arrow. Below the input fields are two buttons: "Login" (dark blue) and "Clear" (light blue). At the bottom of the screen, the text "SDG 6: Clean Water and Sanitation" and "Version 1.0.0" is displayed.

AquaTrace
Water Access & Monitoring System

Login Credentials

Username *
Enter username

Password *
Enter password
Please fill out this field.

Role *
-- Select --

Login **Clear**

SDG 6: Clean Water and Sanitation
Version 1.0.0

Dashboard

The central hub after login displays a summary overview of the entire water system. Provides navigation buttons access to the three main modules, Water Source Management, Maintenance Log and Reports Analytics.

**AquaTrace Dashboard**
Water Access & Monitoring System

Logged in as
Admin (admin)

Logout

System Summary

Total Water Sources
56

Active Sources
45

Pending Maintenance
8

Water Quality Score
87%

Module Navigation


Water Source Management


Maintenance Log


Reports & Analytics

Supporting SDG 6: Ensure access to water and sanitation for all

Sources (Water Sources Management)

A full CRUD (Create, Read, Update, Delete) module for managing water sources. Displays all water sources in a DataGridView table showing details like source name, type (Borehole, Spring, Surface Water), location (Barangay), capacity, status, quality, and last inspection date. Includes search functionality and allows adding, editing, and deleting water source records.

Water Source Management

Back to Dashboard

Search & Actions

Search

Search by any field...

+ New Source

Water Sources Records

ID	Source Name	Type	Location	Capacity	Status	Quality	Last Inspection	Actions
1	Balon ng Barangay San Jose	Borehole	Barangay San Jose	50000 L/day	Active	Good	2026-05-15	 
2	Ilog Pasig Water Intake	Surface Water	Barangay Santa Cruz	100000 L/day	Active	Excellent	2026-05-10	 
3	Bukal ng Barangay Maligaya	Spring	Barangay Maligaya	30000 L/day	Maintenance	Fair	2026-05-01	 

Maintenance (Maintenance Log Management)

Tracks all maintenance activities for water sources. Features a CRUD interface with a DataGridView showing maintenance records. Has a ComboBox that links to water sources from frmSources, so you can select which water source needs maintenance. Records include maintenance type, description, technician name, scheduled/completed dates, status, and cost in Philippine pesos.

Maintenance Log Management

← Back to Dashboard

Search & Actions

Search

Search by any field...

+ New Log

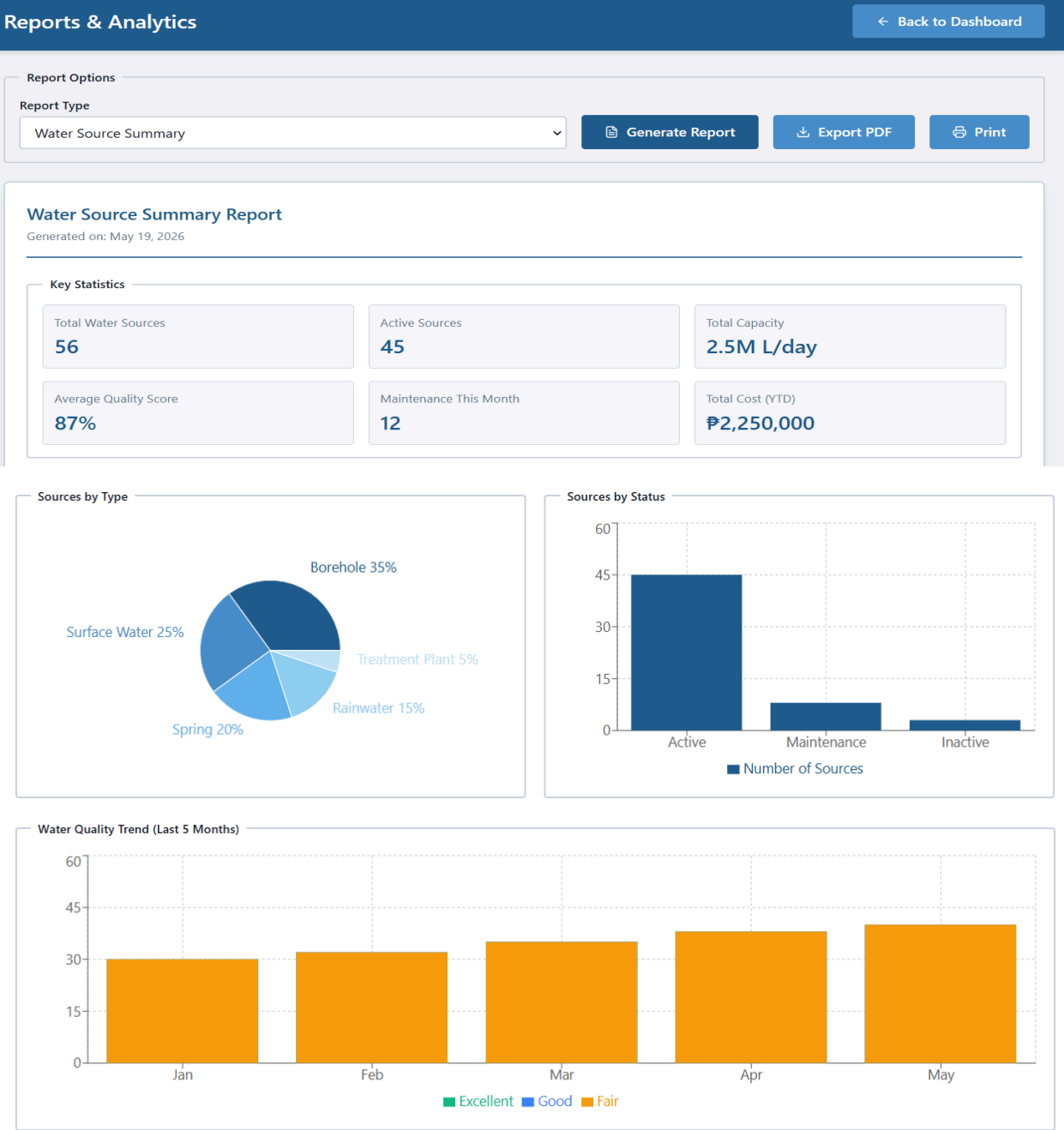
Maintenance Logs

ID	Water Source	Type	Description	Technician	Scheduled	Status	Cost	Actions
1	Balon ng Barangay San Jose	Routine Inspection	Quarterly inspection and pump check	Juan Dela Cruz	2026-05-20	Completed	₱12,500	 
2	Bukal ng Barangay Maligaya	Repair	Fix filtration system leak	Maria Santos	2026-05-18	Pending	₱40,000	 
3	Ilog Pasig Water Intake	Cleaning	Remove debris and sediment	Pedro Reyes	2026-05-15	Completed	₱7,500	 

Reports (Reports Viewer)

Reports Viewer

Displays comprehensive analytics and reports about the water system. Shows visual data through charts including: a pie chart for water source types distribution, bar charts for source status and water quality trends over 5 months. Includes summary statistics and options to generate, export to PDF, or print reports.



11. Individual Contribution

Members	Contributions
Efren V. Taguibulosan	Created the project all class files (ConnectionClass.vb, DBManager.vb, WaterSource.vb, MaintenanceLog.vb, User.vb). Built all 5 Windows Forms (frmLogin,

	frmDashboard, frmSources, frmMaintenance, frmReports), SQL Server database, and documentation.
Mark Perez	Created the Entity Relationship Diagram (ERD) and Data Flow Diagrams (DFD Level 0 and Level 1) using Lucidchart. Contributed to the documentation by writing the SDG Alignment.
John Justin Dela Rosa	Designed the UI/UX mockups and wireframes for all 5 forms using Figma, establishing the color scheme and layout standards used throughout the system.

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